

Section 2 - Practice Problems

Matthew Farah, farahm11@mcmaster.ca, 400468251

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Chapter 1: The Algebraic and Order Properties of $\mathbb{R}$ **Exercise 1.1.1**

Solve the following equations, justifying each step by referring to an appropriate property or theorem.

(a) $2x + 5 = 8$.

Solution:

$$2x + 5 = 8 \implies$$

$$2x + 5 - 5 = 8 - 5 \implies \quad \langle \text{By the additive property of equality} \rangle$$

$$2x + 0 = 3 \implies \quad \langle \text{By the additive inverse property} \rangle$$

$$2x = \implies \quad \langle \text{By the additive identity property} \rangle$$

$$\frac{2x}{2} = \frac{3}{2} \implies \quad \langle \text{By the multiplicative property of equality} \rangle$$

$$\frac{2}{2} \cdot x = \frac{3}{2} \implies \quad \langle \text{By the associativity of multiplication} \rangle$$

$$1 \cdot x = \frac{3}{2} \implies \quad \langle \text{By the multiplicative inverse property} \rangle$$

$$x = \frac{3}{2} \quad \langle \text{By the multiplicative identity property} \rangle$$

(b) $x^2 = 2x$

Solution:

$$x^2 = 2x \implies$$

$$x^2 - 2x = 2x - 2x \implies \langle \text{By the additive property of equality} \rangle$$

$$x^2 - 2x = 0 \implies \langle \text{By the additive inverse property} \rangle$$

$$x \cdot (x - 2) = 0 \implies \langle \text{By the distributive property} \rangle$$

$$x = 0 \text{ or } x = 2 \quad \langle \text{By Theorem 2.1.3.b} \rangle$$

(c) $x^2 - 1 = 3$

Solution:

$$x^2 - 1 = 3 \implies$$

$$x^2 - 1 - 3 = 3 - 3 \implies \langle \text{By the additive property of equality} \rangle$$

$$x^2 - 4 = 0 \implies \langle \text{By the additive inverse property} \rangle$$

$$(x + 2) \cdot (x - 2) = 0 \implies \langle \text{By the distributive property} \rangle$$

$$x = 2 \vee x = -2 \quad \langle \text{By Theorem 2.1.3.a} \rangle$$

(d) $(x - 1)(x - 2) = 0$

Solution:

$$(x - 1)(x - 2) = 0 \implies$$

$$x = 1 \vee x = 2 \quad \langle \text{By Theorem 2.1.3.a} \rangle$$

Exercise 1.1.9

Let $K := \{s + t\sqrt{2} : s, t \in \mathbb{Q}\}$. Show that K satisfies the following:

(a) If $x_1, x_2 \in K$, then $x_1 + x_2 \in K$ and $x_1x_2 \in K$.

Solution:

Let $x_1 = s_1 + t_1\sqrt{2}, x_2 = s_2 + t_2\sqrt{2} \in K$. We separate the proof of each claim into two parts into two parts.

(I) Prove $x_1 + x_2 \in K$

Proof. We can see that:

$$\begin{aligned} x_1 + x_2 &= s_1 + t_1\sqrt{2} + s_2 + t_2\sqrt{2} \\ &= (s_1 + s_2) + (t_1 + t_2)\sqrt{2} \end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under addition, we know that $(s_1 + s_2) \in \mathbb{Q}$ and $(t_1 + t_2) \in \mathbb{Q}$. Therefore, $((s_1 + s_2) + (t_1 + t_2)\sqrt{2}) \in K$ and thus $(x_1 + x_2) \in K$. $\square$

(II) Prove $x_1x_2 \in K$

Proof. We can see that:

$$\begin{aligned}
x_1x_2 &= (s_1 + t_1\sqrt{2})(s_2 + t_2\sqrt{2}) \\
&= s_1s_2 + s_1t_2\sqrt{2} + s_2t_1\sqrt{2} + t_1t_2\sqrt{2}\sqrt{2} \\
&= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + t_1t_2(\sqrt{2})^2 \\
&= s_1s_2 + (s_1t_2 + s_2t_1)\sqrt{2} + 2t_1t_2 \\
&= (s_1s_2 + 2t_1t_2) + (s_1t_2 + s_2t_1)\sqrt{2}
\end{aligned}$$

Thus, since $\mathbb{Q}$ is closed under both addition and multiplication, we know that $s_1s_2 + 2t_1t_2 \in \mathbb{Q}$ and $s_1t_2 + s_2t_1 \in \mathbb{Q}$. Therefore, $(s_1s_2 + 2t_1t_2) + (s_1t_2 + s_2t_1)\sqrt{2} \in K$ and thus $x_1x_2 \in K$. $\square$

(b) If $x \neq 0$ and $x \in K$, then $\frac{1}{x} \in K$.

Solution:

Proof. Let $x = s + t\sqrt{2} \in K$, where s and t cannot equal 0 simultaneously. Therefore, we have:

$$\begin{aligned}
\frac{1}{x} &= \frac{1}{s + t\sqrt{2}} \\
&= \frac{1}{s + t\sqrt{2}} \cdot \frac{s - t\sqrt{2}}{s - t\sqrt{2}} \\
&= \frac{s - t\sqrt{2}}{(s + t\sqrt{2})(s - t\sqrt{2})} \\
&= \frac{s - t\sqrt{2}}{s^2 + (t\sqrt{2})^2} \\
&= \frac{s - t\sqrt{2}}{s^2 + 2t^2} \\
&= \frac{s}{s^2 + 2t^2} - \frac{t}{s^2 + 2t^2}\sqrt{2}
\end{aligned}$$

Therefore, since $\mathbb{Q}$ is closed under addition, multiplication, and division, we know $\frac{s}{s^2 + 2t^2} \in \mathbb{Q}$ and $\frac{t}{s^2 + 2t^2} \in \mathbb{Q}$. Therefore, $\frac{s}{s^2 + 2t^2} - \frac{t}{s^2 + 2t^2}\sqrt{2} \in \mathbb{Q}$ and thus $\frac{1}{x} \in K$ if $x \neq 0$. $\square$

Exercise 1.1.15

If $0 < a < b$, prove that:

(a) $a < \sqrt{ab} < b$

Solution:*Proof.*

$$\begin{aligned}
 ab < b \cdot b &\implies a \cdot a < ab && \implies \\
 ab < b^2 &\implies a^2 < ab && \implies \\
 \sqrt{ab} < \sqrt{b^2} &\implies \sqrt{a^2} < \sqrt{ab} && \implies \\
 \sqrt{ab} < b & \quad a < \sqrt{ab}
 \end{aligned}$$

Therefore, $0 < a < b \implies a < \sqrt{ab} < b$.

□

(b) $\frac{1}{b} < \frac{1}{a}$ **Solution:***Proof.*

$$\begin{aligned}
 a &< b \\
 \frac{a}{1} &< \frac{b}{1} \\
 \frac{a}{1} \cdot 1 &< \frac{b}{1} \cdot 1 \\
 \frac{a}{1} \cdot \frac{b}{b} &< \frac{b}{1} \cdot \frac{a}{a} \\
 \frac{ab}{b} &< \frac{ab}{a} \\
 \frac{ab}{b} \cdot \frac{1}{ab} &< \frac{ab}{a} \cdot \frac{1}{ab} \\
 \frac{1}{b} &< \frac{1}{a}
 \end{aligned}$$

Therefore, $0 < a < b \implies \frac{1}{b} < \frac{1}{a}$.

□

Chapter 2: Absolute Value and the Real Line

Exercise 1.2.9

Find all values of x that satisfy the following inequalities. Sketch all graphs.

(a) $|x - 2| \leq x + 1$

Solution:

$$\begin{aligned} |x - 2| \leq x + 1 & \implies \\ x - 2 \leq x + 1 & \quad x - 2 \leq -(x + 1) \\ & \quad x - 2 \leq -x - 1 \\ & \quad 2x \leq 1 \\ & \quad x \leq \frac{1}{2} \end{aligned}$$

Therefore, the inequality holds whenever $x \in (-\infty, \frac{1}{2}]$.

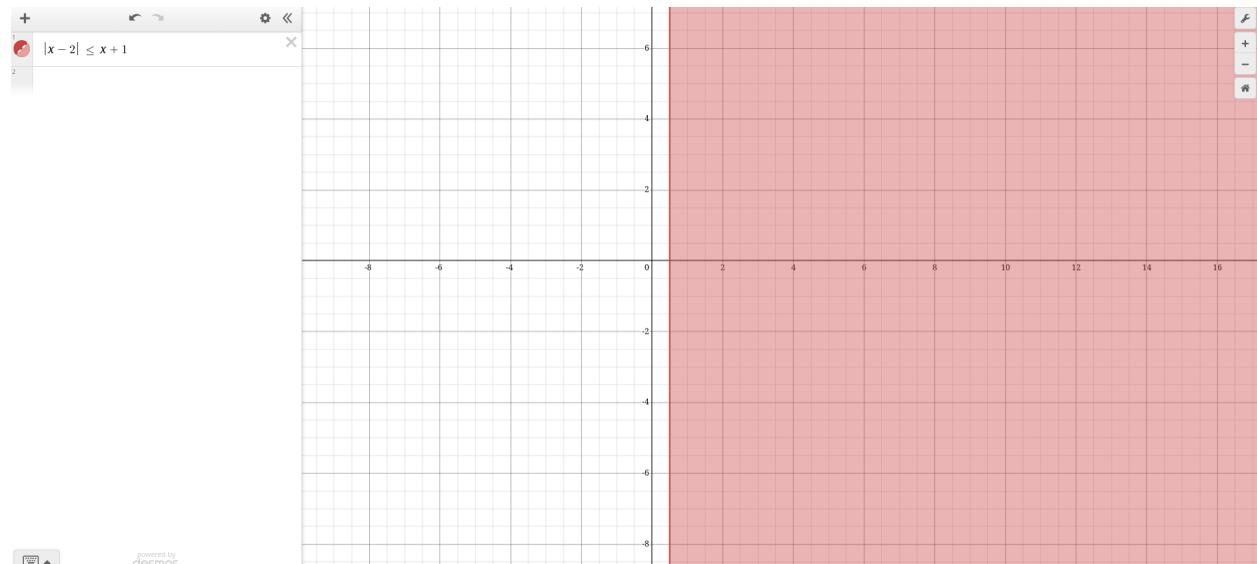


Figure 1: $|x - 2| \leq x + 1$ as graphed by Desmos.

(b) $3|x| \leq 2 - x$

Solution:

$$\begin{array}{llll}
 3|x| & \leq 2 - x & \implies & \\
 3(x) \leq 2 - x & & 3x \leq -(2 - x) & \implies \\
 4x \leq 2 & & 3x \leq 2 + x & \implies \\
 x \leq \frac{1}{2} & & 4x \geq 2 & \\
 & & x \geq \frac{1}{2} &
 \end{array}$$

Therefore, the inequality holds whenever $x \in (-\infty, \frac{1}{2}] \cup [1, \infty)$.

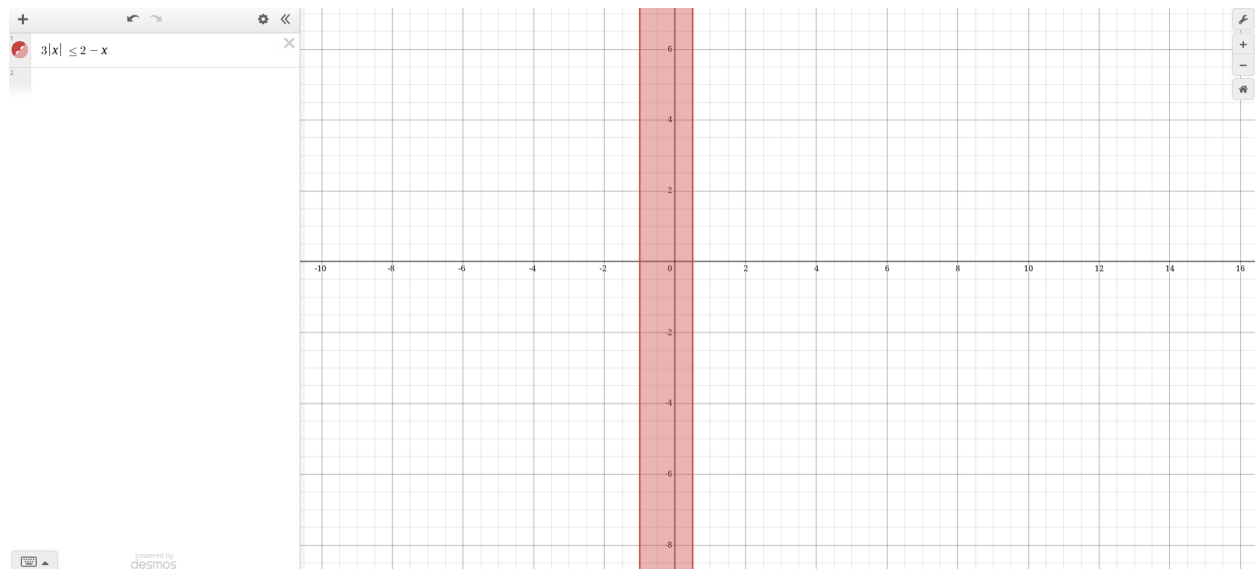


Figure 2: $3|x| \leq 2 - x$ as graphed by Desmos.

Exercise 1.2.14

Determine and sketch the set of pairs $(x, y) \in \mathbb{R} \times \mathbb{R}$ that satisfy:

(a) $|x| = |y|$.

Solution:

$$\begin{aligned}
 |x| = |y| & \implies \\
 (x = y \vee x = -y \vee -x = y \vee -x = -y) & \implies \\
 x = y \vee x = -y &
 \end{aligned}$$

(b) $|x| + |y| = 1$.

(c) $|xy| = 2$.

(d) $|x| - |y| = 2$.

Exercise 1.2.19

Show that if $a, b, c \in \mathbb{R}$, then the “middle number” is $\text{mid}\{a, b, c\} = \min\{\max\{a, b\}, \max\{b, c\}, \max\{c, a\}\}$.

Solution:

Chapter 3: The Completeness Property of $\mathbb{R}$

Exercise 1.3.20

Let $S_3 = \{\frac{1}{n} : n \in \mathbb{N}\}$. Show that $\sup S_3 = 1$ and $\inf S_3 \geq 0$. (It will follow from the Archimedean Property in Section 2.4 that $\inf S_3 = 0$.)

Solution:

(I) Show that $\sup S_3 = 1$

By definition of supremum, we know that 1 is the supremum of S_3 if, for all $\epsilon > 0$:

$$1 \geq s_3 \quad \forall s_3 \in S_3 \text{ AND } \exists s_3 \in S_3 \text{ such that } s_3 > 1 - \epsilon.$$

Without appealing to the archimedean property, we know that 1 must be the supremum of this set because

(II) Show that $\inf s_3 \geq 0$.

Since $n \in \mathbb{N}$, we know $n \geq 0$ by the order limit properties of $\mathbb{R}$ that $\frac{1}{n} \geq 0$ for all $n \in \mathbb{N}$. Now, let's assume for the purposes of contradiction that $s_1 = \inf S_3 < 0$. In such a case, we know that there exists an $s_2 \in \mathbb{R}$ such that $s_1 > s_2 > 0$. However, we know that s_2 must also lower bound S_3 since $\frac{1}{n} > 0 \quad \forall n \in \mathbb{N}$. Thus, any $s_1 > 0$ cannot be the infimum of S_3 since there will always exist a number larger than it (s_2) which lower bounds S_3 . Therefore, by contradiction, the infimum of S_3 must be greater than or equal to zero.

Exercise 1.3.6

Let S be a nonempty subset of $\mathbb{R}$ that is bounded below. Prove that $\inf(S) = -\sup(\{-s : s \in S\})$.

Solution:

Proof. We can appeal to the completeness property of $\mathbb{R}$ to assert that $\inf(S)$ must exist since S is bounded below. Let $m = \inf(S)$. By definition of infimum, we know that for all $s \in S$, $m \leq s$. Therefore, we know that $-m \geq -s$. Therefore, for the set $K = \{-s : s \in S\}$, $-m$ upper bounds K . Now we must prove that $-m$ is the supremum of K . Since m is the infimum of S , we know that $\forall \epsilon > 0, \exists s \in S$ such that $m + \epsilon > s$. Therefore, we have $-m - \epsilon < -s$. Therefore, since m upper bounds K , and for arbitrary $\epsilon > 0$, there always

exists a $k \in K$ such that $-m - \epsilon < k$, $-m$ must be the supremum of $K = \{-s : s \in S\}$. Therefore, appealing to our earlier definition of m , we have:

$$\begin{aligned} m &= m \\ m &= -(-m) \\ \inf(S) &= -\sup(K) \\ \inf(S) &= -\sup(\{-s : s \in S\}) \end{aligned}$$

□

Exercise 1.3.10

Show that if A and B are bounded subsets of $\mathbb{R}$, then $A \cup B$ is a bounded set. Show that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Solution:

We separate our proof into two parts.

(I) Prove that $A \cup B$ is bounded.

Proof. By definition of boundedness, we know that $\exists M_1, M_2 \in \mathbb{R}$ such that $|a| \leq M_1, |b| \leq M_2 \forall a, b \in A, B$. Let $M = \max M_1, M_2$. Thus, we trivially have $M \geq M_1$ and $M \geq M_2$. We know that $\forall c \in (A \cup B), c \in A \vee c \in B$. If $c \in A$, then $|c| \leq M$ since $M \geq M_1$ and if $c \in B$, $|c| \leq M$ since $M \geq M_2$. Thus, $A \cup B$ must be bounded for bounded A, B . □

(II) Prove that $\sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$.

Proof. Appealing to the completeness property of $\mathbb{R}$, since A and B are both bounded, we know that $\sup(A), \sup(B)$ both exist. Let $s_1 = \sup(A)$ and $s_2 = \sup(B)$, and $\epsilon > 0$ be arbitrary. By definition of suprema, we know that $\exists a_0, b_0 \in A, B$:

$$s_1 - \epsilon < a_0 \quad \text{and} \quad s_2 - \epsilon < b_0$$

Now, let $s_3 = \sup(\{s_1, s_2\})$. Since the set $\{s_1, s_2\}$ is finite, we know that the supremum must be an element of the set and thus either $s_3 = s_1$ or $s_3 = s_2$ and therefore $s_3 \geq s_1$ and $s_3 \geq s_2$. Let $c \in A \cup B$. Since $c \in A \vee c \in B$, we know trivially that $c \leq s_3$ meaning s_3 upperbounds $A \cup B$. To show that s_3 must be the supremum, consider again $\epsilon > 0$. From the properties of s_1 and s_2 that we discussed earlier, we know that if $s_3 = s_1$, $\exists a_0 \in A$ such

that $s_3 - \epsilon < a_0$ and if $s_3 = s_2$, $\exists b_0 \in B$ such that $s_3 - \epsilon < b_0$ for all ϵ . Since $a_0, b_0 \in A \cup B$, there always exists a $c_0 \in A \cup B$ such that:

$$s_3 - \epsilon < c_0$$

Thus, by definition of supremum, we know $s_3 = \sup(A \cup B) = \sup(\{\sup(A), \sup(B)\})$. $\square$

Chapter 4: Applications of the Supremum Property

Exercise 1.4.11

If $S := \{\frac{1}{n} - \frac{1}{m} : n, m \in \mathbb{N}\}$, find $\inf(S)$ and $\sup(S)$.

Solution:

I assert that $\inf(S) = -1$ and $\sup(S) = 1$ and my reasoning is as follows:

$$\begin{array}{ccc} 0 < \frac{1}{n} \leq 1, & 0 < \frac{1}{m} \leq 1, & \forall m, n \in \mathbb{N} \implies \\ \frac{1}{n} - \frac{1}{m} \geq 0 - 1 & & \frac{1}{n} - \frac{1}{m} \leq 1 - 0 \\ = -1 & & = 1 \end{array}$$

Therefore, -1 and 1 clearly upper and lowerbound S respectively.

To prove that -1 and 1 are the infimum and supremum of S , consider $\epsilon > 0$. Now consider $1 - \epsilon$. By the Archimedean principle, we know that there exists an $n \in \mathbb{N}$ such that $N > 1 - \epsilon$ and

Chapter 5: Intervals